Homework 1: Data Frames and Visualizations

STAT 141, Week 1

Your name here

Due: Thursday, September 10th at 9:00am on Gradescope as a PDF

In this homework assignment, you will practice:

1. Exploring data frames and their structure
2. Describing and evaluating elements of data visualization
3. Interpreting and comparing examples of the five named graphs
4. Assessing elements of graphs that aid data communication and identifying elements that can be improved

Add any collaborators here!

I collaborated with…

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

When you have completed HW 1, follow this submission process:

- **Step 1**: Export the PDF (not the QMD) of your HW 1 from the RStudio Server. Within the Files tab in the lower right-hand window:
  - Check the box next to "hw01_yourname.pdf".
  - Click on the blue cog and select "Export…" and then "Download."
- **Step 2**: Within Gradescope (which you can access from the course website), click on HW 01 and select the option "Submit PDF."

    - Then "Select PDF" from where it was downloaded onto your computer and hit "Upload PDF."
- **Step 3**: Within the Question Outline, click on "Exercise 1" and then click on the pages that contain your solutions for Exercise 1. Repeat this process for each Problem. **If you do not assign pages to your submission, you will not receive full credit!**
- **Step 4**: Hit "Submit" and revel in the completion of a task!

Exercises

Exercise 1: Syllabus Check-in

We are going to be covering a lot of ground this semester! One of my goals is for the course policies and logistics to be clear and organized, so that we can focus as much of our time and attention on the material as possible.

With this in mind, please read the course syllabus in detail, if you haven't already. Write down three things that stand out to you (they can be positive or negative). They can be related to learning outcomes, policies, content, etc. If you have questions, write them down so that we can clarify.

Exercise 2: Slack Introduction

Slack will be a great place for you to ask and answer questions about course content and (non-urgent) logistics. Asking questions and answering questions from your peers will contribute positively towards your participation grade for the course. I also encourage you to use Slack to connect with your peers from the class to collaborate on assignments together.

For this problem, if you haven't already, please join the course Slack channel (the link can be found on Moodle), and do **one** (or more, if you want) of the following:

- Ask a question about anything from lecture, the lab, or the homework in the corresponding channel
- Give an answer to someone else's question
- Introduce yourself and post a picture of your favorite animal in #animals

Exercise 3: Somerville Data Frame

Let's explore the `somerville` data frame, which is in the `somerville.csv` file.

Starting in 2011, officials from the city of Somerville, Massachusetts decided to start measuring residents' overall happiness and satisfaction with many aspects of life in Somerville; their goal was to become the first city in the United States to systematically track people's happiness. The Happiness Survey is an initiative of the Mayor's Office of Analytics and Innovation (SomerStat) and has the goal of supporting the local government's understanding of what makes Somerville a great place to live.

The happiness survey is sent out every two years to a random sample of Somerville households, with the instruction that each survey should be filled out by one person. The survey asks residents to rate their personal happiness, wellbeing, and satisfaction with City services. This combined dataset includes the survey responses from 2011 to 2019.

Run the code in the following chunk to load the `somerville` data.

```r
somerville <- read.csv("data/somerville.csv")
```

Variables in the `somerville` data:

- `year`: year that response was collected
- `happy`: happiness rating, on a scale from 1 to 10 with 10 being most happy
- `satisfied_life`: rating for satisfaction with life in general, on a scale from 1 to 10 with 10 being most satisfied
- `satisfied_somerville`: rating for satisfaction with Somerville as a place to live, on a scale from 1 to 10 with 10 being most satisfied
- `age`: age in years
- `children_home`: whether respondent has children under 18 living at home
- `ward`: neighborhood ward the respondent lives in
- `precinct`: neighborhood precinct the respondent lives in
- `student`: whether respondent is a student
- `convenient_transport`: rating for how convenient it is to get to where you want to go, on a scale from 1 to 10 with 10 being most satisfied
- `direction_somerville`: whether respondent feels the City is headed in the right direction or is on the wrong track
- `housing`: whether respondent rents or owns their home
- `primary_transport`: primary form of transportation
- `lived_here`: years respondent has lived in Somerville

You can use the `summary()` function on numerical variables to see summary statistics like the minimum and maximum values, and the `count()` function (from the `dplyr` package) to see the number of observations per category for categorical variables. See below for examples of the syntax:

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
summary(somerville$lived_here)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    3.00   10.00   15.89   21.00   91.00
```

```r
count(somerville, primary_transport)
```

```
  primary_transport   n
1              Bike 122
2               Car 460
3    Multiple Modes 524
4    Public transit 281
5              Walk 147
```

Pro tip: Clicking on the name of the data frame in the Environment pane opens a spreadsheet window. In that window, you can sort the data frame by a particular variable by clicking on the arrows in that variable's column.

a. How many observations and variables are in the data frame? What does a row represent in the context of the data?

b. Identify which variables are quantitative and which variables are categorical.

c. Suppose that for the next version of the happiness survey, city officials would like to obtain data on race. Before recent changes, the U.S. Office of Management and Budget required the Census Bureau to have five minimum categories to the question asking respondents for racial identity: White, Black or African American, American Indian or Alaska Native, Asian, and Native Hawaiian or Other Pacific Islander. Discuss the extent to which these categories adequately represent the variable they describe.

d. Choose a categorical variable in the `somerville` data. Discuss the extent to which the given categories adequately represents the variable it describes.

e. Choose a quantitative variable in the `somerville` data, and discuss if the range of values given are reasonable. In other words, think about whether there are values that might be the result of data entry error (such as if someone was listed as being 50 feet tall) and comment on what an unreasonable value might look like.

Exercise 4: Describing and evaluating elements of graphics (Library of Congress)

Check out the graphic entitled "A Brief Visual History of MARC Cataloging at the Library of Congress" using the link. This unique visual was created by Benjamin M. Schmidt so that he could better understand the history of how the Library created their digital card catalogs. The many interesting visual artifacts of the graph also help highlight the significant physical work that was involved in the conversion to digital catalogs.

a. What are the variables in the **data** displayed in this graphic? For each variable, specify if it is categorical or quantitative.

b. How would you describe the **geom**etry being used to display the data?

c. How are each of the variables represented via **aes**thetic elements in the graphic (e.g., x- and y-axis position, color, shape, etc.)?

d. Is other context provided? If so, how does it help the viewer better understand the graphic's story?

e. What does this graph do well?

f. How could this graph be improved?

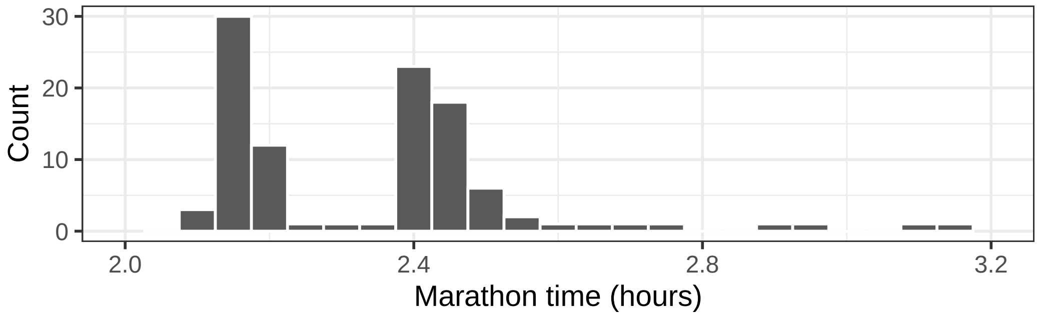
Exercise 5: Interpreting and comparing the five named graphs

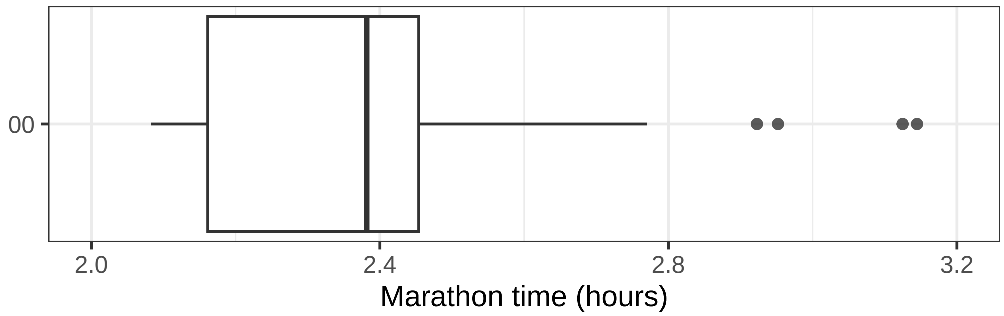
This question is adapted from IMS Chapter 5.

In this problem, we will explore the distribution of finishing times for male and female winners of the New York Marathon between 1970 and 2020. For this problem, you **do not need to edit any code!** Instead, you should click the green, right facing arrow in the code chunks below to get started.

```r
library(openintro)
library(gridExtra)
library(ggplot2)
data(nyc_marathon)
nyc_marathon$division <- factor(nyc_marathon$division,
                                levels = rev(c("Women", "Men")),
                                labels = rev(c("Women's Division",
                                               "Men's Division")))
nyc_marathon <- nyc_marathon[!is.na(nyc_marathon$name), ]

g1 <- ggplot(nyc_marathon, aes(x = time_hrs)) +
  geom_histogram(binwidth = 0.05, color = "white") +
  labs(x = "Marathon time (hours)", y = "Count") +
  scale_x_continuous(limits = c(2, 3.2), breaks = seq(2, 3.2, 0.4)) +
  theme_bw()
g2 <- ggplot(nyc_marathon, aes(x = time_hrs)) +
  geom_boxplot(outlier.alpha = 0.8) +
  labs(x = "Marathon time (hours)", y = "") +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor.y = element_blank(),
        axis.text.y = element_blank()) +
  scale_y_continuous(breaks = c(0),labels = "00")+
  scale_x_continuous(limits = c(2, 3.2), breaks = seq(2, 3.2, 0.4)) +
  theme_bw()
grid.arrange(g1, g2)
```





a. What features of the distribution are apparent in the histogram and not the box plot? What features are apparent in the box plot but not in the histogram?

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b. What may be the reason for the bimodal distribution? Explain.

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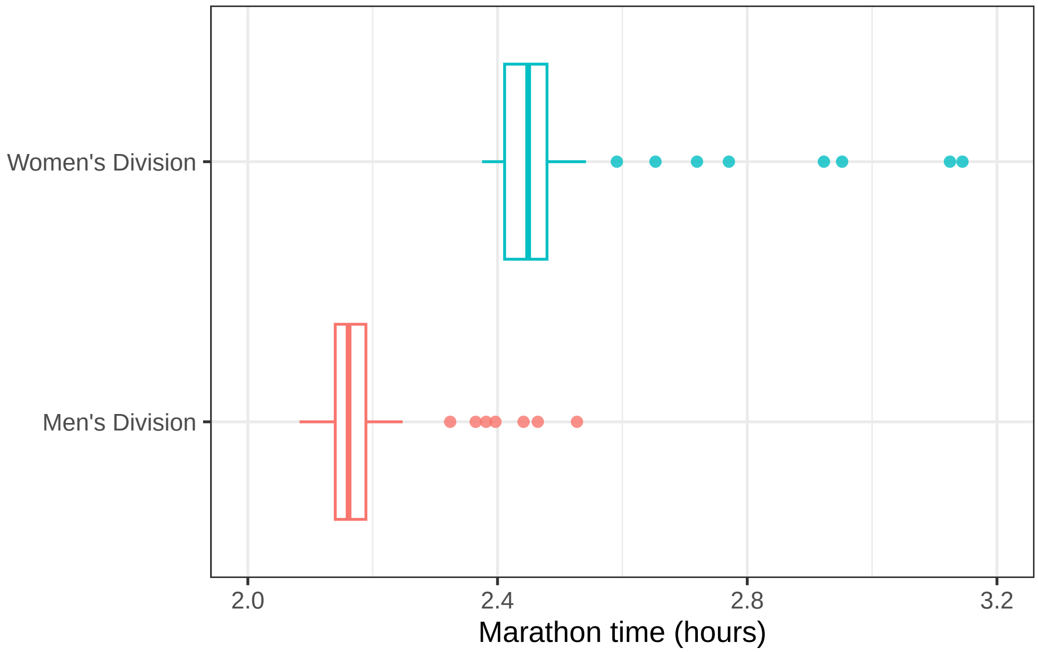
c. How does the box plot shown below help you compare the distribution of marathon times for men and women?

```
ggplot(nyc_marathon, aes(x = time_hrs, y = division, color = division)) +
      geom_boxplot(outlier.alpha = 0.8, show.legend = FALSE) +
      labs(x = "Marathon time (hours)", y = NULL) +
      scale_x_continuous(limits = c(2, 3.2), breaks = seq(2, 3.2, 0.4)) +
  theme_bw()
```



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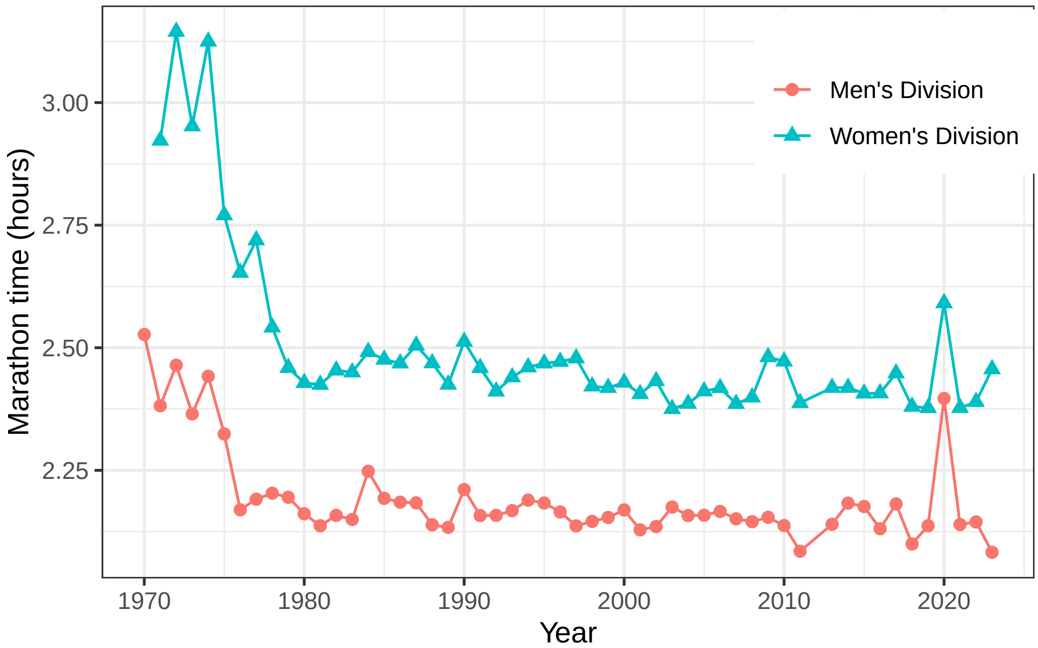
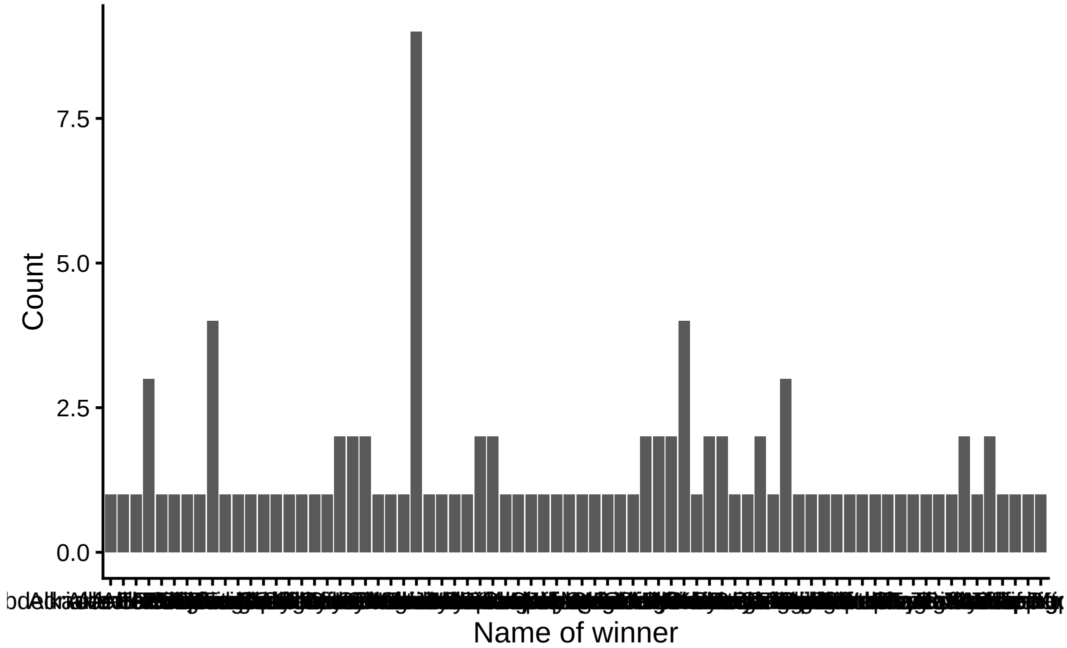
d. The time series plot shown below is another way to look at these data. Describe what is visible in this plot but not in the others.

```
  ggplot(nyc_marathon, aes(x = year, y = time_hrs,
                           color = division, shape = division)) +
    geom_point(size = 2) + geom_line() +
    labs(y = "Marathon time (hours)", x = "Year",color = "",shape = "") +
theme_bw() + theme(legend.position = c(.85,.85))
```



e. The barplot below shows the count of marathon wins per unique name in the data. What insights might this type of plot offer? What issues prevent you from making insights or gaining context?

```
ggplot(nyc_marathon, aes(x = name)) +
  geom_bar() +
  xlab("Name of winner") +
  ylab("Count") +
  theme_classic()
```


7.5
5.0
2.5
0.0
Count
Name of winner